
Spin Hall Effect of Light (SHEL) in Dielectric Metasurfaces

-National Initiative on Undergraduate Science-

NIUS 21.3 Report

Spandan Mandal

Department of Physics , BSc 2023-27

Ramakrishna Mission Vivekananda Centenary College



Project Guide: Dr. Harish N S Krishnamoorthy
TATA Institute of Fundamental Research (TIFR), Hyderabad

Abstract

The Spin Hall Effect of Light (SHEL), a spin-orbit interaction phenomenon in optics, can be significantly influenced by dielectric metasurfaces. In this project, I numerically investigated the SHEL shift using **COMSOL Multiphysics** and **Python**, focusing on amorphous and crystalline Sb_2S_3 -based metastructures. By sweeping wavelength, incident angle, and a structural asymmetry parameter (α), I extracted Fresnel coefficients and computed polarization-resolved SHEL shifts in both transmission and reflection.

Maximum normalized SHEL shifts exceeding 200λ were predicted near normal incidence, although often with very low efficiency. A key result was identifying optimal combinations of α , wavelength, and angle that yield large and efficient SHEL shifts, particularly in the $\alpha = 0$ (symmetric) configuration. These findings provide practical guidelines for future Gaussian beam simulations on 50×50 metasurface arrays, where full parameter scans are computationally expensive. Due to limitations in computing power, direct beam shift simulations could not be completed, but the parameter space explored offers a clear roadmap for further progress.

Acknowledgement

I would like to express my sincere gratitude to **Prof. Harish N. S. Krishnamoorthy** for his invaluable guidance, insightful suggestions, and continuous support throughout the course of this project. His mentorship has been instrumental in shaping my understanding and approach to research.

I am especially thankful to **Abel Paul Joseph**, PhD student at TIFR Hyderabad, for his patient and detailed help during the initial stages of simulation work in COMSOL. His technical support greatly eased my transition into working with complex simulation environments.

I also extend my thanks to all members of the research group and senior students at TIFR Hyderabad for their encouragement, discussions, and assistance that contributed significantly to my learning.

I am grateful to the **NIUS Physics Program coordinators at HBCSE-TIFR** for providing me the opportunity to be involved in this intellectually stimulating project and for creating such a supportive research environment.

Finally, I thank my friends and peers, whose encouragement and moral support helped me stay motivated and focused throughout the work, and my **parents** for their constant love, patience, and support behind the scenes.

Contents

Abstract	1
Acknowledgement	2
1 Introduction	1
2 Theory	3
2.1 Fresnel Coefficients and Polarization-Dependent Phase Shifts	3
2.2 Geometric Phase and the Poincaré Sphere	4
2.3 Spin Hall Shift from Field Distribution and Analytical Approximation	6
3 Methodology	9
3.1 Plane-Wave Analysis	9
3.2 Realistic Beam Simulation	11
4 Results and Discussion	14
4.1 Polarization-Dependent Optical Response	14
4.2 Wavelength-Dependent SHEL Shift	14
4.3 Material-Dependent SHEL Contrast	16
4.4 Angular Dependence of SHEL Shift	17
4.5 Reflectance-Based SHEL Shift	17
5 Conclusion	20
A	21
A.1 Meta-Atom Geometry for Different Asymmetry Parameters	21
A.2 Additional Wavelength-Dependent SHEL Spectra	22
A.3 Additional Material-Dependent SHEL Contrast Spectra	22
A.4 Additional Angular Dependence of SHEL Shift	22
A.5 Additional Reflectance-Based SHEL Shift	22

Chapter 1

Introduction

Spin-orbit interaction, the coupling between the spin (intrinsic angular momentum) and trajectory (orbital motion) of a particle or wave, has become an important concept across classical mechanics, quantum mechanics, and especially optics.

In optics, spin-orbit coupling arises from angular momentum conservation. A change in one form of angular momentum (spin or orbital) must be accompanied by a compensating change in the other, so that the total angular momentum remains conserved.

One of the most striking manifestations of this in optics is the **Spin Hall Effect of Light (SHEL)**, a transverse spatial shift between right and left circularly polarized components of light when a beam reflects or refracts at an interface.

When a linearly polarized light beam (a coherent superposition of right and left circularly polarized components - RCP and LCP) is incident on a planar optical interface, the reflected or refracted beam experiences a transverse shift perpendicular to the plane of incidence. This shift is spin-dependent, leading to a separation of RCP and LCP components. The SHEL is therefore a purely optical analog of the Spin Hall Effect in condensed matter systems, governed by spin-orbit interactions in momentum space.

Beyond the SHEL, several other spin-dependent optical effects have attracted significant attention in modern photonics, such as the **optical Magnus effect**, **Pancharatnam–Berry phase-induced angular splitting**, and related **geometric phase phenomena** [1]. While each effect is distinct in origin and manifestation, they all emerge from the underlying spin-orbit interaction of light.

In the context of the SHEL, this transverse spin-dependent shift can be understood through the geometric (Berry) phase [2] associated with polarization evolution on the Poincaré sphere or in momentum space. Metasurfaces, which are engineered subwavelength arrays of nanostructures, provide a powerful platform to manipulate such geometric phases. They enable precise control over light's amplitude, phase, and polarization, and thus can be designed to enhance or tailor the SHEL shift by introducing abrupt phase discontinuities at subwavelength scales.

In recent years, the Spin Hall Effect of Light (SHEL) has been observed at various types of interfaces, including air-dielectric, air-metal, and air-semiconductor boundaries. However, the SHEL shifts in these systems are typically very small, often on the subwavelength scale, and become significant only under oblique incidence. Experimental detection of such shifts, especially the Imbert-Fedorov (I-F) shift, is challenging due to their small magnitude.

Interestingly, at the Brewster angle, the Fresnel reflection coefficient for p-polarized light (r_p) becomes zero, while the coefficient for s-polarized light (r_s) remains nonzero. This leads to strong SHEL shifts at Brewster's angle - but the overall reflection efficiency becomes very low, making practical detection difficult.

SHEL is now finding applications in optical sensing and detector design. Its sensitivity to optical and material parameters allows it to be used in measuring film thickness [3, 4],

refractive index [5], magnetic properties [6], optical constants, molecular concentrations, and even humidity.

Recent advances have shown dynamic tunability of SHEL using graphene [7], where applying electric or magnetic fields allows active control of the beam shift. Additionally, researchers have proposed polarization-independent SHEL, where engineered structures with equal Fresnel coefficients for s and p polarizations can produce symmetric shifts even with unpolarized light.

Specifically, I used amorphous and crystalline Sb_2S_3 as dielectric layers in metastructure models and investigated how SHEL shifts change by varying symmetry-breaking parameters, wavelengths, and incident angles. I simulated both s- and p-polarized light inputs and focused mainly on near-normal incidence, ranging from 0.5° to 10° .

In my simulations, the maximum SHEL shift achieved was about 175λ ($\lambda = 680$ nm), with an efficiency of approximately 90%, using s- and p-polarized plane waves incident at 0.1° on a crystalline Sb_2S_3 interface.

Chapter 2

Theory

The theoretical framework underlying this work is based on established studies in the literature, including the comprehensive 2023 review on the Spin Hall Effect of Light (SHEL) [1], along with several investigations on metasurface-enhanced spin-optical phenomena [8, 9, 10, 11, 12]. Although the present work does not introduce new theoretical developments, the analysis is grounded in a well-defined conceptual framework involving polarization-dependent Fresnel reflection, the emergence of geometric phase, and their role in the Spin Hall Effect of Light.

Enhancement of SHEL shifts using dielectric metasurfaces, particularly through structural symmetry breaking and resonance engineering, forms the central motivation for the numerical investigations carried out in this study. This section outlines the fundamental physical principles governing these mechanisms, with emphasis on how phase shifts, symmetry properties, and topological effects give rise to transverse spin-dependent beam shifts.

2.1 Fresnel Coefficients and Polarization-Dependent Phase Shifts

The Fresnel reflection and transmission coefficients provide the fundamental theoretical framework for analyzing polarization-dependent optical phenomena at dielectric interfaces [13, 14]. These coefficients describe both the amplitude and phase response of the *s*- (perpendicular) and *p*- (parallel) polarized components when an electromagnetic wave is incident on the boundary between two dielectric media.

For an interface separating media with refractive indices n_1 and n_2 , and an incident angle θ_i , the Fresnel reflection coefficients are expressed as

$$r_s = \frac{n_1 \cos \theta_i - n_2 \cos \theta_t}{n_1 \cos \theta_i + n_2 \cos \theta_t}, \quad (2.1)$$

$$r_p = \frac{n_2 \cos \theta_i - n_1 \cos \theta_t}{n_2 \cos \theta_i + n_1 \cos \theta_t}, \quad (2.2)$$

where θ_t denotes the transmission angle determined from Snell's law,

$$n_1 \sin \theta_i = n_2 \sin \theta_t. \quad (2.3)$$

The corresponding Fresnel transmission coefficients are given by

$$t_s = \frac{2n_1 \cos \theta_i}{n_1 \cos \theta_i + n_2 \cos \theta_t}, \quad (2.4)$$

$$t_p = \frac{2n_1 \cos \theta_i}{n_2 \cos \theta_i + n_1 \cos \theta_t}. \quad (2.5)$$

In general, these coefficients are complex-valued and therefore introduce polarization-dependent phase shifts upon reflection and transmission. Even when the incident light is linearly polarized, the differential phase accumulation between the s - and p -polarized components can modify the polarization state of the reflected or transmitted field, often producing elliptical polarization.

The relative phase difference between the reflected polarization components is defined as

$$\phi_R = \arg(r_p) - \arg(r_s), \quad (2.6)$$

where $\arg(\cdot)$ denotes the argument of the complex reflection coefficient. This phase difference plays a crucial role in the emergence of transverse spin-dependent beam shifts associated with the Spin Hall Effect of Light.

Similarly, the phase difference introduced during transmission can be written as

$$\phi_T = \arg(t_p) - \arg(t_s). \quad (2.7)$$

These polarization-dependent phase differences generate phase gradients across the angular spectrum of the beam. As illustrated in Fig. 2.1, such phase gradients can lead to transverse displacements of the beam centroid in real space.

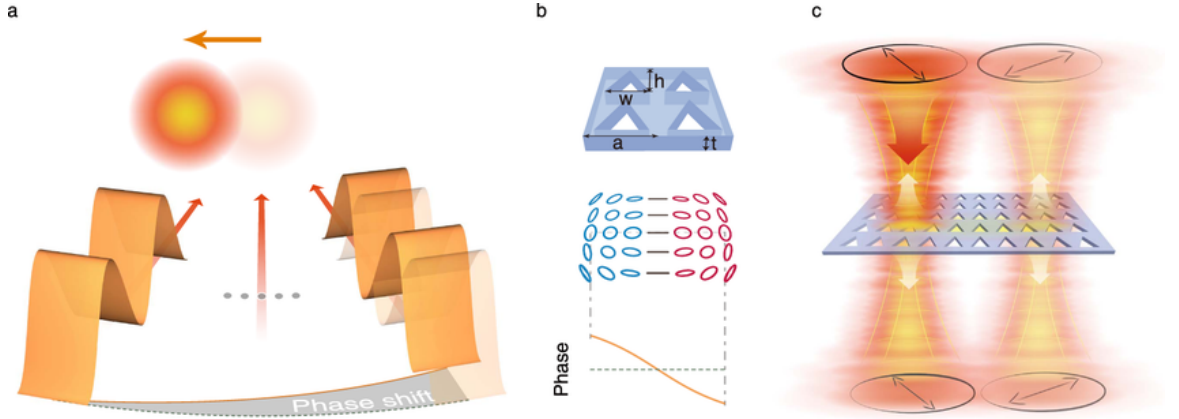


Figure 2.1: Conceptual illustration of polarization-dependent beam shifts arising from geometric phase gradients in momentum space. A spatial variation of the geometric phase across different wave-vector components of a beam leads to a transverse displacement of the beam centroid in real space. Adapted from Ref. [11].

In metasurface-based systems, subwavelength structuring enables precise control over these polarization-dependent phase responses. By tailoring the geometry and arrangement of meta-atoms, it becomes possible to engineer the amplitude and phase of the Fresnel coefficients, thereby enhancing spin-dependent beam shifts. This capability forms the theoretical foundation for the numerical analysis and metasurface designs investigated in this work [1, 8].

2.2 Geometric Phase and the Poincaré Sphere

The geometric phase, also known as the Pancharatnam–Berry (PB) phase, plays a fundamental role in polarization-dependent optical phenomena. Unlike the dynamical phase, which originates from the optical path length accumulated during propagation, the geometric phase arises from the evolution of the polarization state of light and depends solely on the trajectory traced by the polarization state on the Poincaré sphere [12].

When an optical field undergoes a transformation in its polarization state—for example during propagation, reflection, or refraction—the final polarization state may acquire an additional phase relative to the initial state even if both states share identical intensity and polarization characteristics. According to Pancharatnam’s criterion, two polarization states are considered in phase when their interference produces maximum intensity. Any phase difference required to satisfy this condition can therefore be interpreted as a geometric phase associated with the evolution of the polarization state.

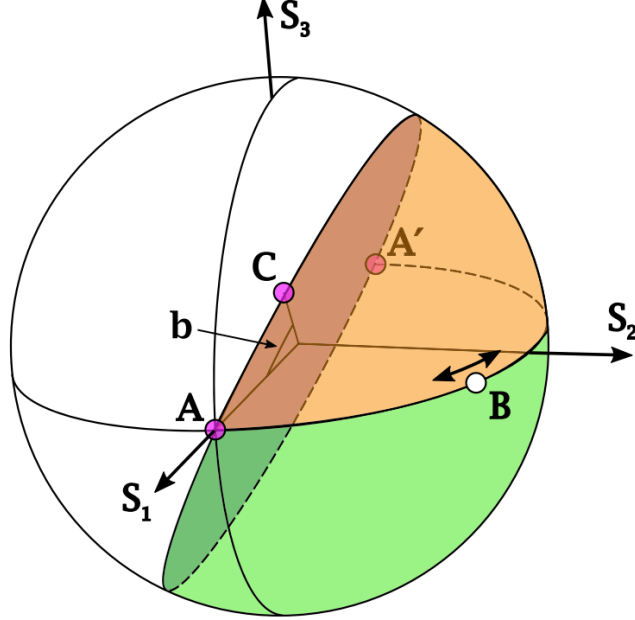


Figure 2.2: Representation of polarization states on the Poincaré sphere. Each point on the sphere corresponds to a distinct state of polarization, and the geometric phase accumulated during polarization evolution is related to the solid angle enclosed by the trajectory on the sphere. Adapted from Ref. [12].

As illustrated in Fig. 2.2, polarization states can be represented as points on the Poincaré sphere, where different trajectories correspond to different polarization transformations. When the polarization state evolves along a closed path on the sphere, the accumulated geometric phase is determined by the solid angle subtended by that trajectory.

For a polarization state evolving along a closed path on the Poincaré sphere, the accumulated geometric phase is given by

$$\Phi_g = -\frac{1}{2}\Omega, \quad (2.8)$$

where Ω denotes the solid angle subtended by the closed trajectory at the center of the Poincaré sphere. This phase is intrinsically spin-dependent and is closely related to the intrinsic angular momentum carried by the polarization state of the optical field.

In metasurfaces exhibiting spatially varying optical anisotropy, the geometric phase can be directly encoded through the orientation of individual unit cells. Rotating anisotropic nanostructures such as elliptical pillars, nanorods, or nano-apertures modifies the local polarization response of the metasurface and introduces a phase delay that depends on both the orientation angle of the structure and the helicity of the incident light [8, 1]. As a result, a spatially varying orientation of meta-atoms generates a spin-dependent phase gradient across the interface.

This spin-dependent geometric phase gradient produces a transverse displacement of the optical beam through spin–orbit interaction of light, which constitutes one of the fundamental

mechanisms responsible for the Spin Hall Effect of Light in metasurface-based systems. In momentum-space formulations, particularly when considering transverse wave-vector components such as k_y , the geometric phase becomes explicitly helicity dependent and can be expressed as

$$\Phi_G = \sigma \Phi, \quad (2.9)$$

where $\sigma = \pm 1$ represents the helicity corresponding to left- and right-circular polarizations, and Φ denotes the accumulated geometric phase associated with the polarization transformation. This helicity-dependent phase modulation leads to opposite phase gradients for opposite spin states and ultimately results in the spin-dependent transverse beam shifts characteristic of the Spin Hall Effect of Light.

2.3 Spin Hall Shift from Field Distribution and Analytical Approximation

In numerical simulations and experimental measurements, the Spin Hall Effect of Light (SHEL) is commonly quantified through the transverse displacement of the beam centroid. When a polarized beam interacts with an optical interface, the left- and right-circular polarization components experience opposite transverse shifts due to spin-orbit interaction. This results in a spin-dependent displacement perpendicular to the plane of incidence.

For a transmitted or reflected beam, the transverse shift can therefore be defined as the centroid position of the optical intensity distribution along the y -direction. For left- and right-circular polarizations ($\sigma = \pm 1$), the SHEL shift for horizontally and vertically polarized incidence can be written as

$$\delta_H^\pm = \frac{\iint y |\tilde{E}_{t,H}^\pm(x_t, y_t)|^2 dx_t dy_t}{\iint |\tilde{E}_{t,H}^\pm(x_t, y_t)|^2 dx_t dy_t}, \quad (2.10)$$

$$\delta_V^\pm = \frac{\iint y |\tilde{E}_{t,V}^\pm(x_t, y_t)|^2 dx_t dy_t}{\iint |\tilde{E}_{t,V}^\pm(x_t, y_t)|^2 dx_t dy_t}. \quad (2.11)$$

Here $\tilde{E}_{t,H}^\pm$ and $\tilde{E}_{t,V}^\pm$ denote the transmitted electric-field components for horizontal (H) and vertical (V) polarization states, respectively. These centroid definitions directly relate the SHEL shift to the spatial distribution of the optical field and therefore provide a practical method for extracting beam displacement from simulated or measured intensity profiles.

Analytical expressions for the SHEL shift can be derived using the angular spectrum representation of a Gaussian beam combined with the Fresnel transmission coefficients of the interface. For a paraxial Gaussian beam with beam waist w_0 incident at an angle θ_i , the transverse shifts for the two orthogonal polarization states can be expressed as

$$\delta_H^\pm = \pm \frac{k_0 w_0^2 \cot \theta_i (|t_p|^2 - \text{Re}(t_p t_s^*))}{k_0^2 w_0^2 |t_p|^2 + |t_p - t_s|^2 \cot^2 \theta_i + \left(\frac{dt_p}{d\theta_i} \frac{dt_p^*}{d\theta_i} \right)}, \quad (2.12)$$

$$\delta_V^\pm = \pm \frac{k_0 w_0^2 \cot \theta_i (|t_s|^2 - \text{Re}(t_s t_p^*))}{k_0^2 w_0^2 |t_s|^2 + |t_s - t_p|^2 \cot^2 \theta_i + \left(\frac{dt_s}{d\theta_i} \frac{dt_s^*}{d\theta_i} \right)}, \quad (2.13)$$

where $k_0 = 2\pi/\lambda$ is the free-space wave number and t_s and t_p are the Fresnel transmission coefficients for s - and p -polarized light. These expressions explicitly demonstrate that the

magnitude of the SHEL shift depends on both the amplitude and phase relations between the polarization-dependent transmission coefficients.

For most practical beam configurations, the weak-focusing condition

$$k_0^2 w_0^2 \gg \cot^2 \theta_i \quad (2.14)$$

is satisfied. Under this approximation, the analytical expressions simplify considerably, yielding

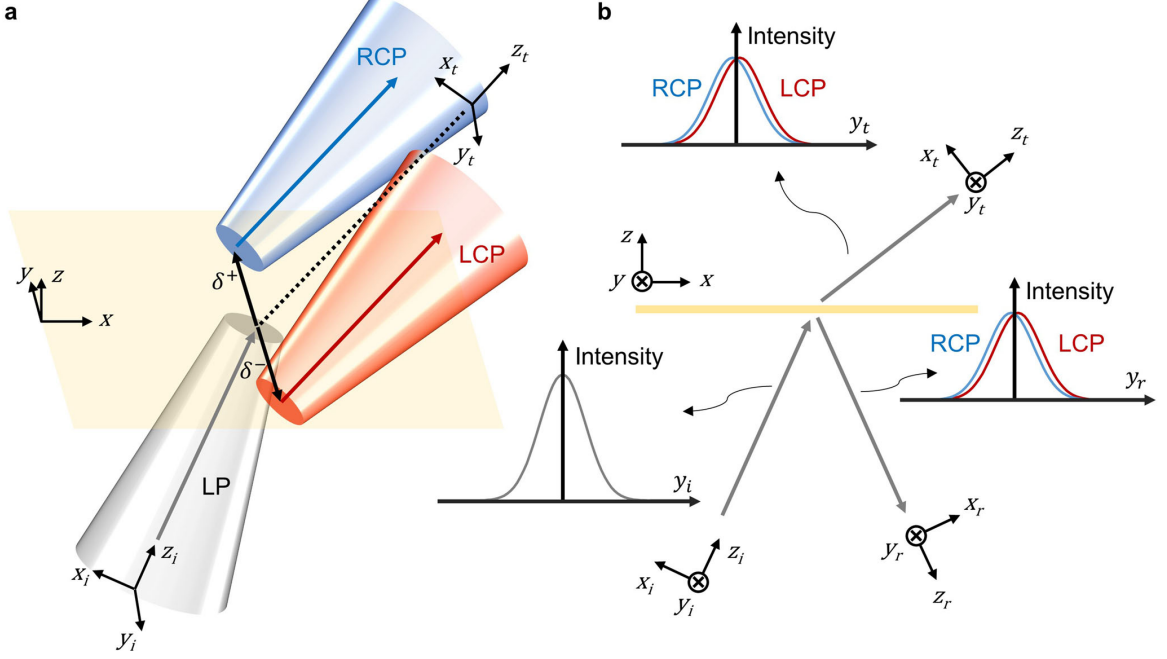


Figure 2.3: Schematic illustration of the Spin Hall Effect of Light (SHEL). A linearly polarized beam incident at an optical interface splits into two opposite circular polarization components (LCP and RCP). Due to spin-orbit interaction of light, the two components undergo opposite transverse displacements along the y -direction, resulting in a spin-dependent beam shift $\delta^\pm$. Adapted from Ref. [1].

$$\delta_H^\pm \approx \pm \frac{\cot \theta_i}{k_0} \operatorname{Re} \left(1 - \frac{t_s}{t_p} \right), \quad (2.15)$$

$$\delta_V^\pm \approx \pm \frac{\cot \theta_i}{k_0} \operatorname{Re} \left(1 - \frac{t_p}{t_s} \right). \quad (2.16)$$

These simplified forms highlight that the SHEL shift is governed by the complex ratio of the polarization-dependent transmission coefficients. In particular, the relative phase difference between the two polarizations plays a crucial role in determining the magnitude of the transverse displacement.

An alternative formulation expresses the shift explicitly in terms of the amplitude ratio and phase difference between the two transmission coefficients [8]:

$$\delta_H^\pm = \pm \frac{\lambda}{2\pi} \cot \theta_i \left(1 - \frac{|t_s|}{|t_p|} \cos \phi \right), \quad (2.17)$$

$$\delta_V^\pm = \pm \frac{\lambda}{2\pi} \cot \theta_i \left(1 - \frac{|t_p|}{|t_s|} \cos \phi \right), \quad (2.18)$$

where

$$\phi = \arg(t_s) - \arg(t_p) \quad (2.19)$$

is the relative phase between the two transmission coefficients. These relations show that large SHEL shifts may arise either from strong phase differences between the two polarization channels or from significant asymmetry in their transmission amplitudes.

The validity of the above approximations requires

$$\left(\frac{2\pi w_0}{\lambda}\right)^2 \gg \cot^2 \theta_i, \quad (2.20)$$

which may break down for tightly focused beams or near-normal incidence. In this regime the paraxial approximation becomes invalid, and the system may transition from a simple transverse spin Hall shift to more complex phenomena such as optical vortex generation and spin-to-orbital angular momentum conversion [8].

A fundamental limitation in SHEL enhancement arises from the intrinsic trade-off between transverse shift magnitude and optical efficiency. Increasing the shift typically requires reducing the magnitude of one of the transmission coefficients, which simultaneously decreases the transmitted optical power. The corresponding transmission efficiencies for the two polarization states are given by

$$\epsilon_H = |t_p|^2, \quad (2.21)$$

$$\epsilon_V = |t_s|^2. \quad (2.22)$$

Therefore, practical metasurface designs aim to engineer large polarization-dependent phase differences while maintaining high transmission amplitudes. Structures operating near quasi-bound states in the continuum (quasi-BICs) provide a promising platform for achieving this balance, enabling both large SHEL shifts and high optical efficiency in dielectric metasurfaces.

Chapter 3

Methodology

The SHEL shift formulas derived from Fresnel coefficients are strictly valid for plane waves, as these coefficients are obtained under the assumption of infinite, uniform wavefronts. In realistic experimental conditions, however, light is typically delivered in the form of Gaussian beams possessing finite transverse extent and curved wavefronts. Consequently, direct application of plane-wave Fresnel theory represents an approximation, and deviations may arise when evaluating beam shifts.

To address this distinction, the investigation was divided into two complementary components:

1. **Plane-wave analysis:** Linearly polarized plane waves were used as excitation on a single meta-cell to extract Fresnel coefficients and predict the maximum achievable SHEL shift under idealized conditions.
2. **Realistic beam simulation:** A Gaussian polarized beam incident on a 50×50 metasurface array was simulated to model experimentally relevant conditions and to evaluate the manifestation of SHEL at the full-interface level.

3.1 Plane-Wave Analysis

An asymmetry parameter α [9] was introduced to quantify structural dissymmetry, defined as

$$\alpha = \frac{\Delta L}{L}.$$

Here, $\alpha = 0$ corresponds to a perfect square geometry, while $\alpha = 1$ corresponds to a bilaterally triangular configuration (Fig A.1). The parameter ΔL represents the change in the length of one side while maintaining a constant base dimension.

The numerical simulations were performed in **COMSOL Multiphysics** using a three-dimensional model under the *Electromagnetic Waves, Frequency Domain (ewfd)* interface with a frequency-domain study. The unit meta-cell consisted of a quartz (glass) substrate of thickness $16t_0$, above which a dielectric layer of amorphous or crystalline Sb_2S_3 with thickness $t_0 = 100$ nm was placed. The structure contained a hole of constant area $S_0 = 207 \times 207$ nm², with periodicity $a = 377$ nm. An air column of thickness $16t_0$ was placed above the dielectric layer.

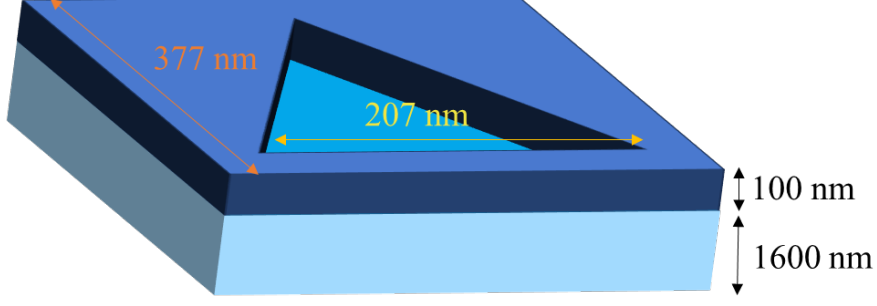


Figure 3.1: Metaphotonic unit cell with dissymmetry parameter $\alpha = 1$.

The geometric parameters were defined as

$$L = \sqrt{\frac{2S_0}{(2-\alpha)}},$$

$$W = \begin{cases} L - \alpha, & \alpha < 1 \\ 0, & \alpha \geq 1 \end{cases}$$

Floquet periodic boundary conditions were applied in the x and y directions. Port conditions were assigned along the $\pm z$ directions, with linearly polarized s- or p-polarized excitation incident at an angle θ_i in the x - z plane.

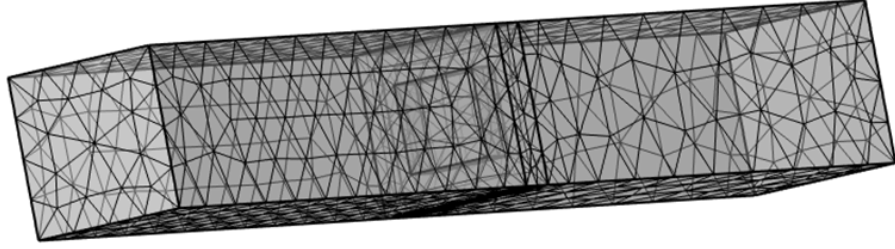


Figure 3.2: Meshing configuration used in COMSOL simulations.

Two independent parametric sweeps were performed:

1. Wavelength sweep from 370 nm to 1110 nm at fixed $\theta_i = 5^\circ$.
2. Incident-angle sweep from 0.5° to 10° at fixed wavelength $\lambda = 680$ nm.

Wavelength Sweep

The asymmetry parameter α was varied over the set $\{1, 0.75, 0.50, 0.25, 0.10, 0\}$. For each value, a wavelength sweep from 370 nm to 1110 nm in steps of 5 nm was performed at fixed $\theta_i = 5^\circ$. Fresnel reflection and transmission coefficients were extracted for each configuration. For each case, transmittance, reflectance, and associated phase shifts for s- and p-polarized components were computed:

$$\begin{aligned}
T_s &= \text{S-polarized transmittance}, \\
R_s &= \text{S-polarized reflectance}, \\
T_p &= \text{P-polarized transmittance}, \\
R_p &= \text{P-polarized reflectance}, \\
\phi_\lambda^T &= \arg T_p - \arg T_s, \\
\phi_\lambda^R &= \arg R_p - \arg R_s.
\end{aligned}$$

The computed data were exported and post-processed using Python to evaluate phase difference, intensity variation, and the corresponding SHEL shifts. Transmission efficiency was simultaneously assessed. A comparative study was conducted between amorphous and crystalline Sb_2S_3 to quantify material-dependent variations in SHEL magnitude.

Angle Sweep

Using the same set of asymmetry parameters, an angular sweep from 0.5° to 10° in increments of 0.1° was performed at fixed wavelength $\lambda = 680$ nm. Fresnel coefficients were extracted as functions of incident angle, and the resulting SHEL shifts and transmission efficiencies were evaluated for each configuration.

3.2 Realistic Beam Simulation

A Gaussian beam propagating along the z -axis is described by the scalar electric field distribution

$$E(x, y, z) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{x^2 + y^2}{w^2(z)}\right) \exp\left[-i\left(kz + \frac{k(x^2 + y^2)}{2R(z)} - \psi(z)\right)\right], \quad (3.1)$$

where E_0 is the peak electric field amplitude at the beam waist ($z = 0$), w_0 is the beam waist radius, $k = 2\pi/\lambda$ is the wave number, and λ is the wavelength. The Rayleigh range is given by

$$z_R = \frac{\pi w_0^2}{\lambda}, \quad (3.2)$$

with the beam radius

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2}, \quad (3.3)$$

the radius of curvature

$$R(z) = z \left[1 + \left(\frac{z_R}{z}\right)^2\right], \quad (3.4)$$

and the Gouy phase shift

$$\psi(z) = \tan^{-1}\left(\frac{z}{z_R}\right). \quad (3.5)$$

The corresponding intensity distribution is

$$I(x, y, z) = I_0 \left(\frac{w_0}{w(z)} \right)^2 \exp \left(-\frac{2(x^2 + y^2)}{w^2(z)} \right). \quad (3.6)$$

For a vector Gaussian beam linearly polarized along $\hat{x}$,

$$\vec{E}(x, y, z) = \hat{x} E(x, y, z), \quad (3.7)$$

while for circular polarization relevant to spin-orbit interactions,

$$\vec{E}(x, y, z) = \frac{1}{\sqrt{2}}(\hat{x} \pm i\hat{y})E(x, y, z). \quad (3.8)$$

Under the paraxial approximation, longitudinal field components are negligible for moderately focused beams. At the beam waist ($z = 0$), the field components can therefore be approximated as follows:

S-polarized (TE) light:

$$E_x = 0, \quad (3.9)$$

$$E_y = E_0 \exp \left(-\frac{x^2 + y^2}{w_0^2} \right), \quad (3.10)$$

$$E_z = 0. \quad (3.11)$$

P-polarized (TM) light:

$$E_x = E_0 \exp \left(-\frac{x^2 + y^2}{w_0^2} \right), \quad (3.12)$$

$$E_y = 0, \quad (3.13)$$

$$E_z = 0. \quad (3.14)$$

For an obliquely incident Gaussian beam with incident angle θ_i in the x - z plane, a rotated coordinate system (x', y', z') is introduced such that propagation occurs along z' . The coordinate transformation is

$$x' = x \cos \theta_i + z \sin \theta_i, \quad (3.15)$$

$$y' = y, \quad (3.16)$$

$$z' = z \cos \theta_i - x \sin \theta_i. \quad (3.17)$$

In this rotated frame, the transverse envelope at $z' = 0$ becomes

$$E_0 \exp \left(-\frac{(x \cos \theta_i + z \sin \theta_i)^2 + y^2}{w_0^2} \right). \quad (3.18)$$

The electric field components in the laboratory frame are therefore expressed as:

S-polarized (TE) light:

$$E_x = 0, \quad (3.19)$$

$$E_y = E_0 \exp \left(-\frac{(x \cos \theta_i + z \sin \theta_i)^2 + y^2}{w_0^2} \right), \quad (3.20)$$

$$E_z = 0. \quad (3.21)$$

P-polarized (TM) light:

$$E_x = E_0 \cos \theta_i \exp \left(-\frac{(x \cos \theta_i + z \sin \theta_i)^2 + y^2}{w_0^2} \right), \quad (3.22)$$

$$E_y = 0, \quad (3.23)$$

$$E_z = E_0 \sin \theta_i \exp \left(-\frac{(x \cos \theta_i + z \sin \theta_i)^2 + y^2}{w_0^2} \right). \quad (3.24)$$

A metasurface array consisting of 50×50 unit cells was modeled to approximate realistic experimental conditions. Scattering Boundary Conditions (SBCs) were applied along the transverse (x and y) directions to emulate open boundaries and prevent artificial lateral reflections. Perfectly Matched Layers (PMLs) were implemented along the $\pm z$ directions to absorb outgoing radiation and ensure accurate modeling of free-space propagation.

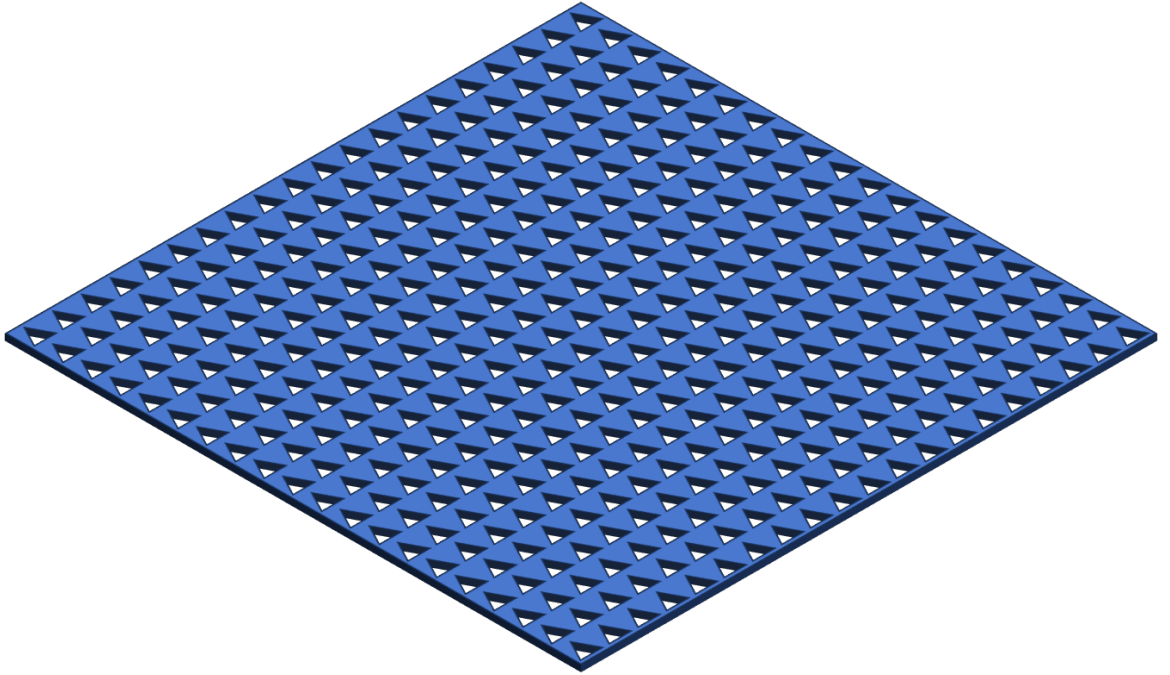


Figure 3.3: Schematic of the triangular nanopatch metasurface composed of subwavelength apertures arranged periodically.

For excitation, a user-defined input port was employed to prescribe the spatially localized Gaussian field profile analytically, enabling control over beam waist, polarization state, and angle of incidence. The output port was configured as an analyzer to resolve polarization-dependent transmission components, allowing investigation of spin-dependent effects. Direct full-wave simulation of the SHEL shift in both real and momentum space for the complete array proved computationally intensive within the available resources. Consequently, quantitative beam-shift extraction from this configuration remains an open extension for future investigation.

Chapter 4

Results and Discussion

This section presents the results obtained from numerical simulations performed using *COM-SOL Multiphysics* and subsequent data analysis using *Python*. The primary objective of this analysis was to evaluate the magnitude and efficiency of the Spin Hall Effect of Light (SHEL) in metasurfaces composed of amorphous and crystalline Sb_2S_3 .

The simulations were carried out by systematically varying several parameters, including the incident wavelength, the angle of incidence, and the structural asymmetry parameter α .

For each configuration, the transmittance, reflectance, and phase information for both *s*- and *p*-polarized light were extracted. These quantities were then used to calculate the resulting SHEL shift and its corresponding optical efficiency.

The goal of this analysis was to identify configurations where large SHEL shifts occur together with reasonably high optical efficiency, which is crucial for practical observation and applications.

4.1 Polarization-Dependent Optical Response

The polarization-dependent optical response of the metasurface for the maximally asymmetric case ($\alpha = 1$) is shown in Fig. 4.1.

The figure presents the wavelength dependence of the transmittance and reflectance spectra for both *s*- and *p*-polarized light for amorphous and crystalline Sb_2S_3 .

At $\alpha = 1$, the metasurface exhibits strong polarization-dependent resonant behavior. In particular, the Fano resonance peak for *p*-polarized light is stronger than that for *s*-polarized light. However, as the asymmetry parameter α decreases, the resonance strength shifts and the *s*-polarized resonance becomes dominant.

For example, in the case of amorphous Sb_2S_3 , the reflectance Fano peak in R_s increases when α decreases from 1 to 0.5, but then decreases again when $\alpha = 0.25$.

A pronounced dip in reflectance is observed near 690 nm, which likely corresponds to a strong resonant condition caused by mode coupling or destructive interference. Additionally, a clear Fano resonance feature appears around 740 nm.

In contrast, the crystalline metasurface shows a red-shifted resonance, indicating that the optical response strongly depends on the material phase.

These resonant features play a crucial role in determining the magnitude of the resulting SHEL shifts.

4.2 Wavelength-Dependent SHEL Shift

The wavelength dependence of the SHEL shift for $\alpha = 0.25$ is shown in Fig. A.2.

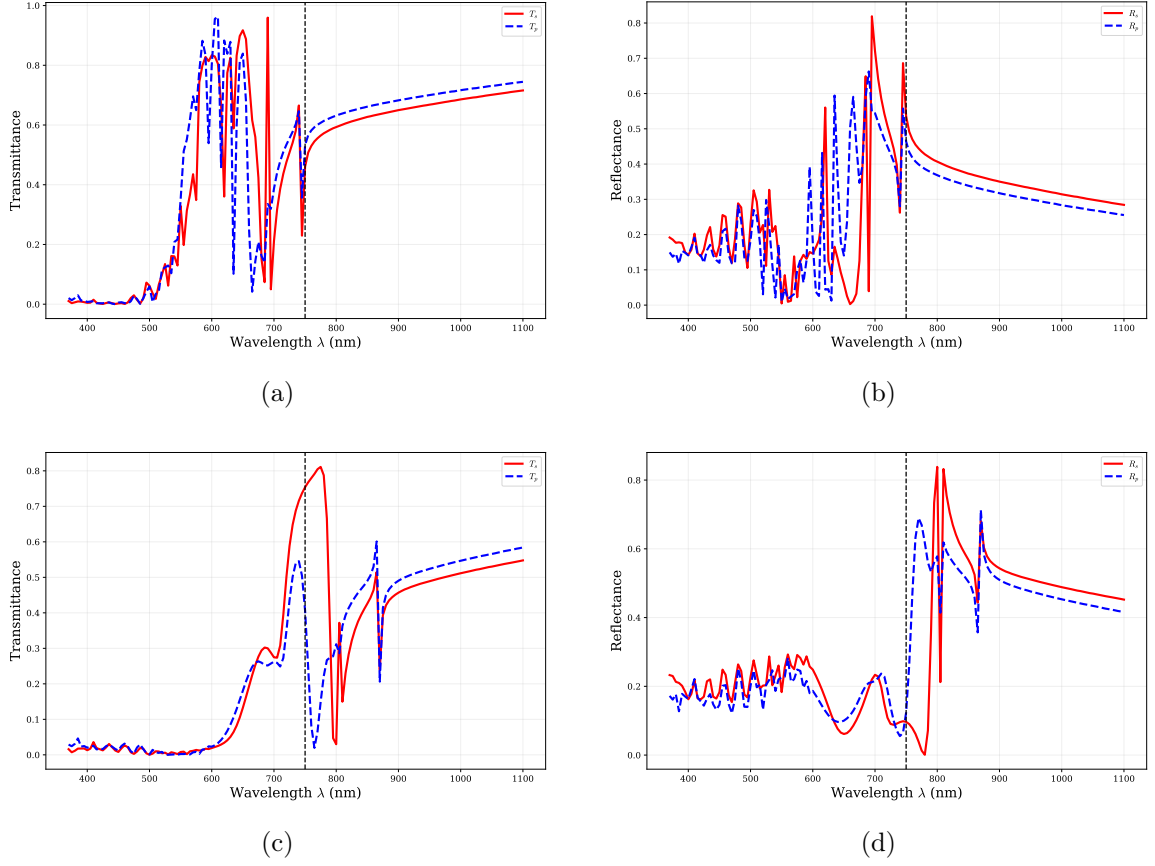


Figure 4.1: Polarization-dependent optical response of Sb_2S_3 metasurfaces at $\alpha = 1$. (a) Transmittance spectrum of the amorphous phase showing T_s and T_p . (b) Reflectance spectrum of the amorphous phase showing R_s and R_p . (c) Transmittance spectrum of the crystalline phase. (d) Reflectance spectrum of the crystalline phase. The dashed vertical line marks the reference wavelength of 750 nm.

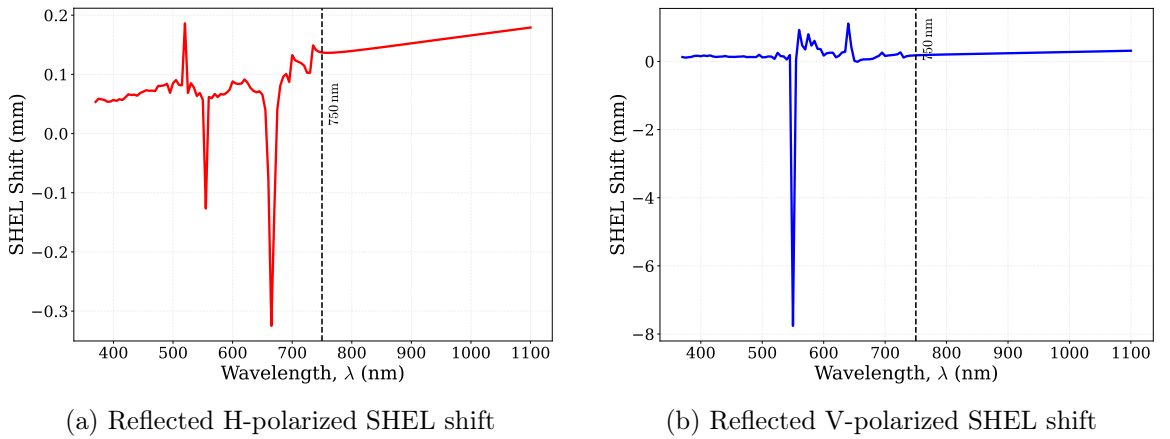


Figure 4.2: Wavelength dependence of the Spin Hall Effect of Light (SHEL) shift for an amorphous Sb_2S_3 metasurface at $\alpha = 0.25$ and an incident angle of 5° . (a) SHEL shift for horizontally polarized reflected light. (b) SHEL shift for vertically polarized reflected light. A strong resonance-enhanced spin-dependent beam displacement appears near the visible wavelength range around 545–555 nm.

This figure compares the SHEL shift spectra for amorphous and crystalline Sb_2S_3 metasurfaces under both horizontal (H) and vertical (V) polarization for transmitted and reflected beams.

The most striking feature occurs in the reflected V-polarized channel for the amorphous structure, where a sharp negative shift exceeding 8 mm appears around 550 nm (Fig. 4.2). This large shift arises from strong spin-orbit coupling associated with the resonant optical response of the metasurface.

In contrast, the crystalline metasurface exhibits significantly smaller and smoother shifts, typically remaining below 1.00 mm across the entire spectral range.

These results demonstrate that the material phase strongly influences the spin-orbit interaction of light in the metasurface.

Large peaks in the SHEL shift appear at wavelengths where the transmittance or reflectance approaches zero. This behavior follows directly from the analytical expression of the SHEL shift, which contains ratios of Fresnel coefficients.

Specifically,

- when T_s or R_s approaches zero, the V-polarized SHEL shift diverges;
- when T_p or R_p approaches zero, the H-polarized SHEL shift diverges.

However, despite the large predicted shifts, the corresponding optical efficiency becomes extremely small, typically within the range 0–5%. As a result, these large theoretical shifts may be difficult to observe experimentally.

4.3 Material-Dependent SHEL Contrast

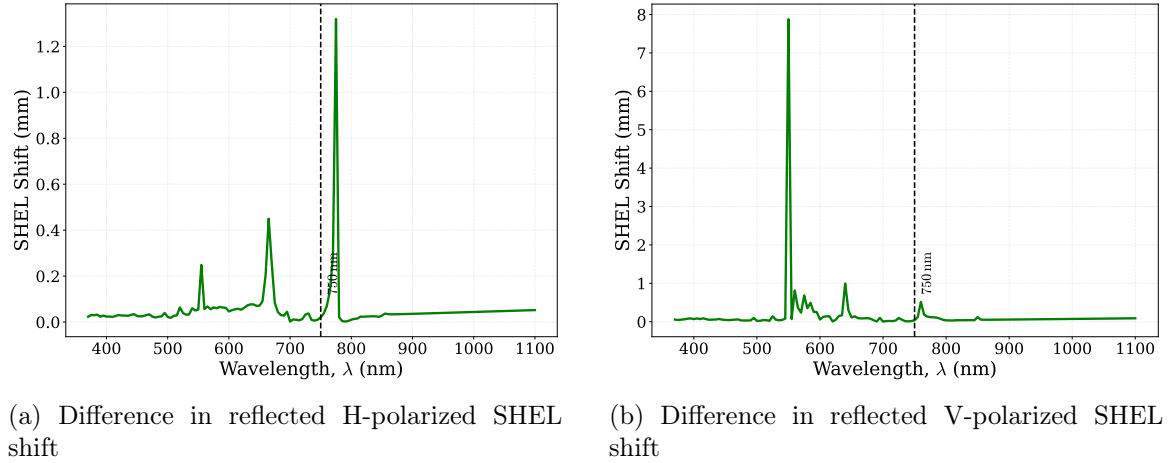


Figure 4.3: Difference in the Spin Hall Effect of Light (SHEL) shift between amorphous and crystalline Sb_2S_3 metasurfaces as a function of wavelength for $\alpha = 0.25$ and an incident angle of 5° .

To quantify the effect of the material phase on the SHEL response, the absolute difference in SHEL shift magnitude

$$|\Delta| = |\delta_{\text{amor}} - \delta_{\text{cryst}}| \quad (4.1)$$

was calculated for each polarization configuration.

The resulting spectra are shown in Fig. A.3.

The largest contrast appears in the reflected V-polarized channel, where the shift difference reaches nearly 8 mm near 550 nm 4.3(b). This large difference originates from the stronger resonance in the amorphous structure compared to the crystalline one.

Additional sharp features are also observed in other channels between 670 nm and 775 nm.

Despite these large differences, the associated transmission and reflection efficiencies remain extremely low, typically below 1%. Consequently, these large spectral contrasts are mainly useful for understanding the underlying physics of spin-dependent light-matter interaction rather than for direct applications.

4.4 Angular Dependence of SHEL Shift

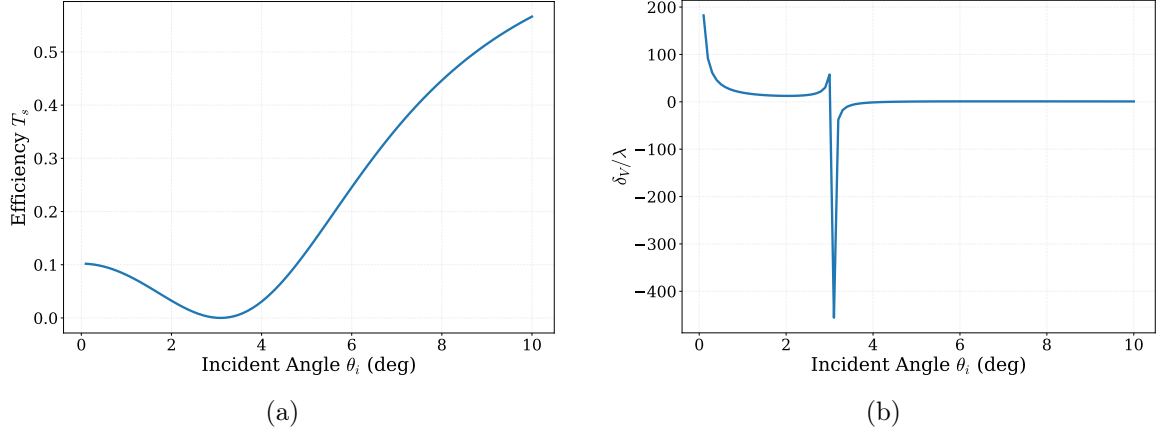


Figure 4.4: Angular dependence of the normalized Spin Hall Effect of Light (SHEL) shift and transmission efficiency for V-polarized light in the amorphous Sb_2S_3 metasurface at $\alpha = 0$ and $\lambda = 680$ nm. (a) Transmission efficiency as a function of incident angle. (b) Corresponding normalized SHEL shift.

To further examine the behavior of the SHEL shift, the incident angle was varied from 0.1° to 10° at a fixed wavelength of 680 nm.

The resulting transmission-based SHEL shifts and corresponding efficiencies are shown in Fig. A.4.

According to the analytical theory of the SHEL shift, the shift diverges as the incident angle approaches normal incidence ($\theta \rightarrow 0^\circ$). This trend is consistent with the numerical results.

In the amorphous metasurface, the normalized SHEL shift for H polarization exceeds 200λ at incident angles below 0.1° , while for V polarization the shift reaches approximately 400λ near 3° , as shown in Fig. 4.4. However, these extremely large shifts occur in angular regions where the transmission efficiency is significantly reduced, typically below 10% and in some cases approaching zero.

These results indicate an inherent trade-off between the magnitude of the SHEL shift and the corresponding transmitted optical power.

4.5 Reflectance-Based SHEL Shift

The angular dependence of the SHEL shift in reflection is shown in Fig. A.5.

At very small incident angles near 0.1° , both amorphous and crystalline metasurfaces exhibit large normalized shifts of approximately 175λ .

However, the corresponding reflectance efficiency shows a strong dependence on the material phase. The amorphous metasurface maintains a high efficiency of nearly 90%, whereas the crystalline structure exhibits an efficiency of approximately 10%.

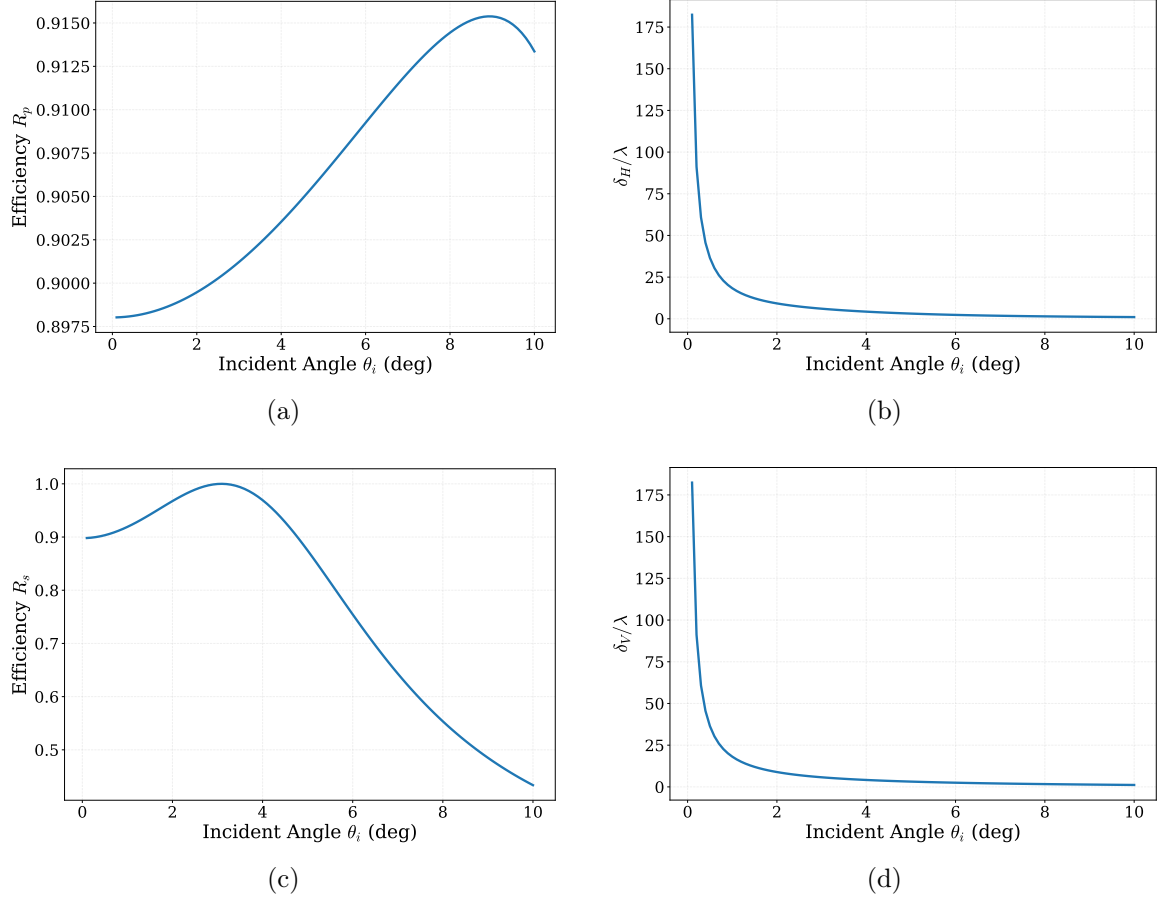


Figure 4.5: Angular dependence of the normalized Spin Hall Effect of Light (SHEL) shift and reflectance efficiency for the amorphous Sb_2S_3 metasurface at $\alpha = 0$ and $\lambda = 680$ nm. (a) Reflectance efficiency for H polarization. (b) Normalized SHEL shift for H polarization. (c) Reflectance efficiency for V polarization. (d) Normalized SHEL shift for V polarization.

At a more moderate incident angle of about 3° , the shift magnitude decreases to approximately $5\text{--}10\lambda$ for both materials, as shown in Fig. 4.5. In this regime, the amorphous metasurface maintains nearly 100% reflectance efficiency, while the crystalline structure remains limited to around 10%.

These results indicate that amorphous Sb_2S_3 metasurfaces provide a more favorable balance between large SHEL shift and high optical efficiency, suggesting greater suitability for practical implementations.

Chapter 5

Conclusion

In this project, I numerically investigated the Spin Hall Effect of Light (SHEL) in dielectric metasurfaces composed of amorphous and crystalline Sb_2S_3 , using COMSOL Multiphysics for simulation and Python for post-processing and analysis.

The SHEL shift was explored under both transmission and reflection geometries, for s- and p-polarized light, by sweeping parameters such as wavelength, angle of incidence, and a symmetry-breaking parameter α . The phase differences between Fresnel coefficients for the two polarizations were used to calculate the spin-dependent lateral beam shifts.

Key findings of the study include:

- Extremely large SHEL shifts (up to $\sim 8 \text{ nm}$) were observed near resonant conditions for amorphous Sb_2S_3 , particularly in reflected V-polarized light around 550 nm. However, these occurred in regions of very low optical efficiency (0-1%), limiting their practical usability.
- The crystalline counterpart exhibited smoother, lower-amplitude shifts across all channels, but often with better spectral confinement.
- Angular sweep studies showed that near-normal incidence ($\theta \approx 0.1^\circ$) leads to diverging SHEL shift values ($\sim 175\lambda$), consistent with the theoretical prediction. However, efficiency considerations are crucial, the amorphous metasurface showed up to 90% reflectance efficiency in this regime, while the crystalline sample remained below 10%.
- Among all configurations, $\alpha = 0$ (fully symmetric unit cell) provided the most optimal trade-off between large SHEL shift and high transmission or reflection efficiency, particularly for amorphous Sb_2S_3 .
- Comparisons of SHEL shift spectra and their differences between amorphous and crystalline metasurfaces helped isolate the role of material phase on spin-orbit interaction and geometric phase accumulation.

Overall, this project demonstrates how metasurface symmetry, material phase, and input polarization collectively influence the magnitude and detectability of SHEL shifts. The results serve as a practical guideline for identifying optimal combinations of wavelength, symmetry-breaking parameter (α), and incident angle that are most likely to exhibit pronounced SHEL shifts. This insight is especially valuable when attempting to simulate the effect directly using a Gaussian beam on a full 50×50 metasurface array, where exhaustive parameter scanning is computationally infeasible. Although I initiated the Gaussian beam simulations in COMSOL, the model could not yield conclusive results due to the high computational load. Further refinement and optimization of the model are needed to make progress in this direction.

Appendix A

A.1 Meta-Atom Geometry for Different Asymmetry Parameters

The metasurface unit-cell geometry used in the simulations is controlled by the asymmetry parameter α . As α varies from 0 to 1, the aperture shape evolves from a symmetric square to a triangular structure, gradually breaking the in-plane symmetry of the meta-atom.

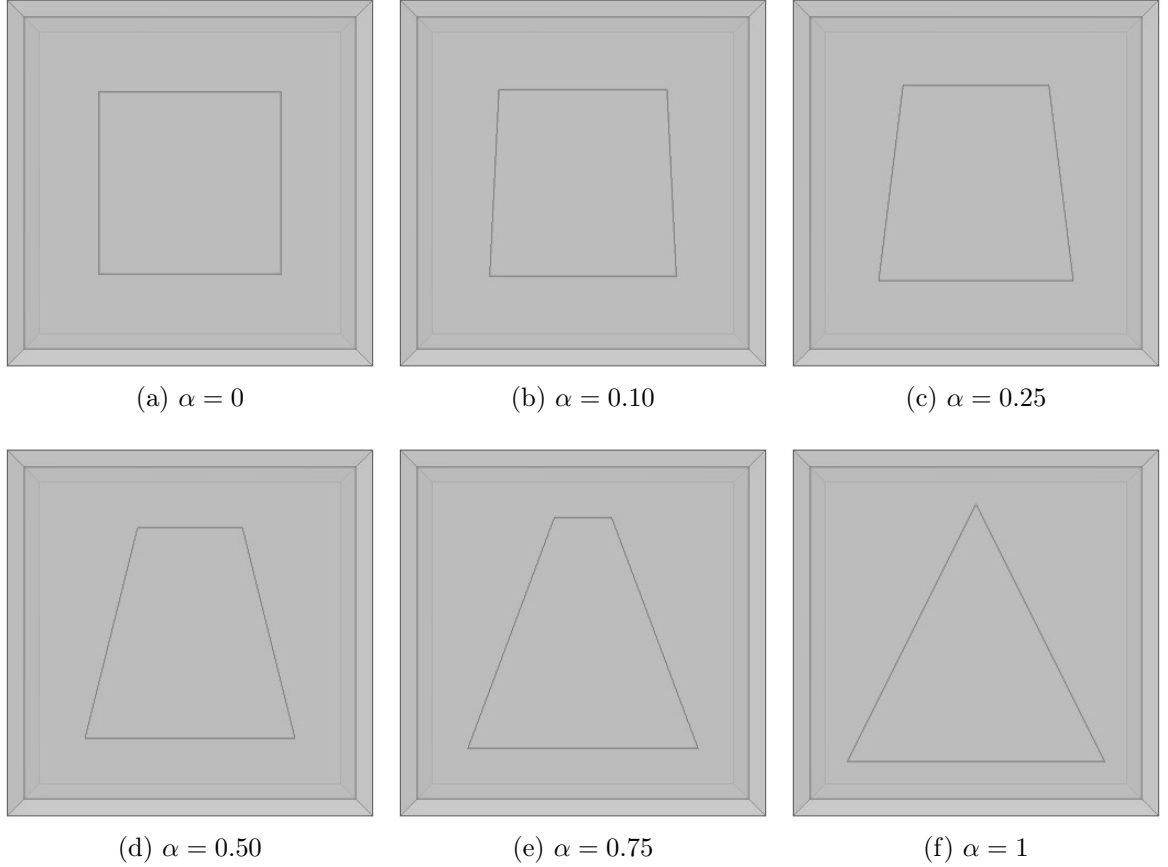


Figure A.1: Meta-atom geometries used in the metasurface design for different values of the asymmetry parameter α . The parameter α controls the degree of symmetry breaking of the aperture shape inside the unit cell. When $\alpha = 0$, the structure is perfectly symmetric (square aperture). As α increases, the geometry gradually deforms, and at $\alpha = 1$ the aperture becomes triangular, representing the maximally asymmetric configuration.

A.2 Additional Wavelength-Dependent SHEL Spectra

For completeness, this appendix presents the remaining wavelength-dependent SHEL spectra for $\alpha = 0.25$ that were not included in the main text.

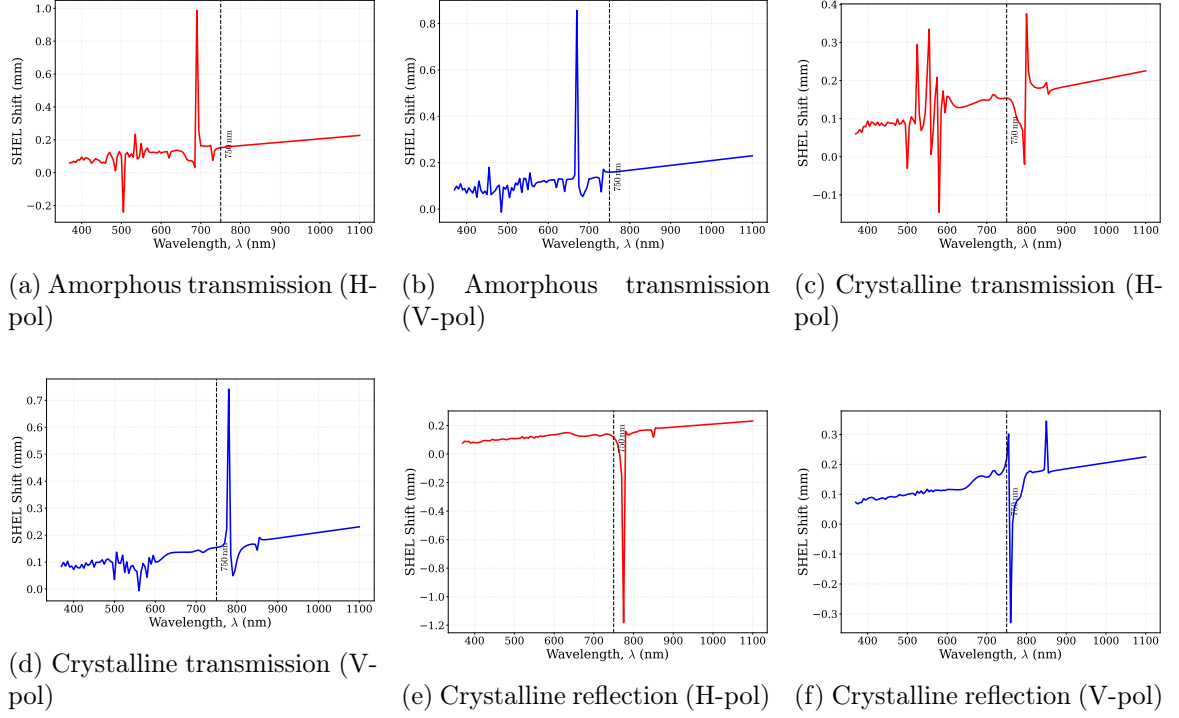


Figure A.2: Additional wavelength-dependent SHEL shift spectra for Sb_2S_3 metasurfaces at $\alpha = 0.25$ and an incident angle of 5° . Panels (a,b) correspond to transmission from the amorphous metasurface, (c,d) show transmission from the crystalline metasurface, and (e,f) represent reflection from the crystalline structure. The dashed vertical line indicates the reference wavelength $\lambda = 750$ nm.

A.3 Additional Material-Dependent SHEL Contrast Spectra

For completeness, the remaining wavelength-dependent SHEL shift differences between amorphous and crystalline Sb_2S_3 metasurfaces are presented here for $\alpha = 0.25$ and an incident angle of 5° . These figures correspond to the transmission channels that were not included in the main text.

A.4 Additional Angular Dependence of SHEL Shift

For completeness, the remaining angular-dependent SHEL shift and transmission efficiency spectra for $\alpha = 0$ are presented here. These figures correspond to the polarization channels and material configurations that were not included in the main text.

A.5 Additional Reflectance-Based SHEL Shift

For completeness, the angular dependence of the SHEL shift and reflectance efficiency for the crystalline Sb_2S_3 metasurface is presented here for $\alpha = 0$ and $\lambda = 680$ nm. These figures correspond to the polarization channels that were not included in the main text.

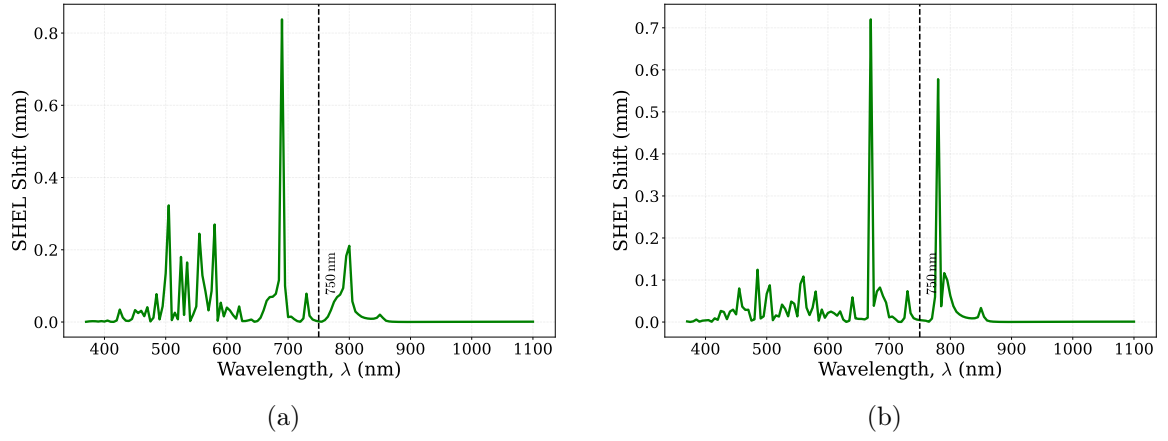


Figure A.3: Additional wavelength-dependent difference in SHEL shift magnitude between amorphous and crystalline Sb_2S_3 metasurfaces. (a) Difference in transmitted H-polarized SHEL shift. (b) Difference in transmitted V-polarized SHEL shift. The dashed vertical line marks the reference wavelength $\lambda = 750$ nm.

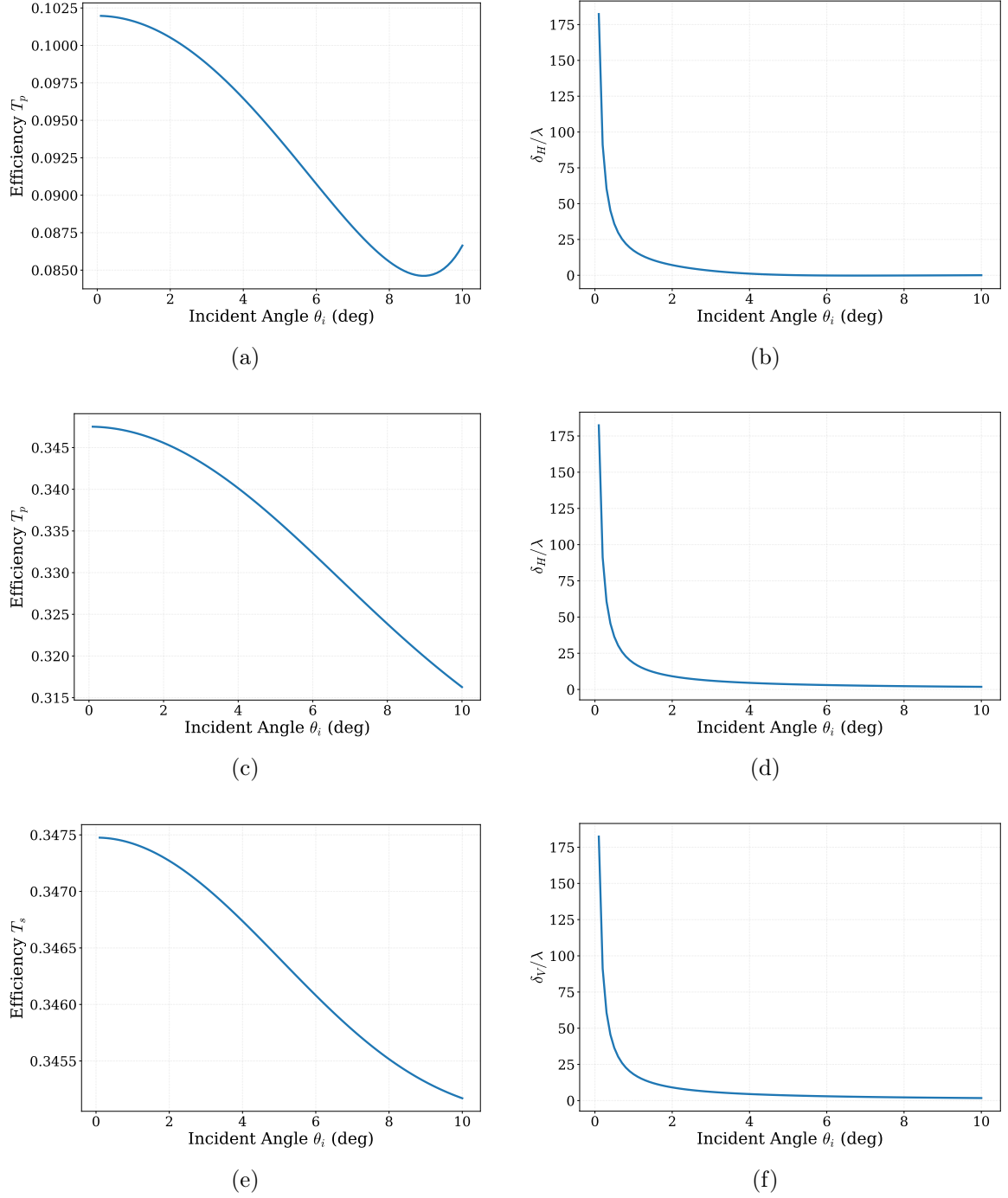


Figure A.4: Additional angular dependence of the normalized SHEL shift and transmission efficiency for Sb_2S_3 metasurfaces at $\alpha = 0$ and $\lambda = 680 \text{ nm}$. Panels (a,b) correspond to the amorphous metasurface for H polarization, (c,d) represent the crystalline metasurface for H polarization, and (e,f) correspond to the crystalline metasurface for V polarization.

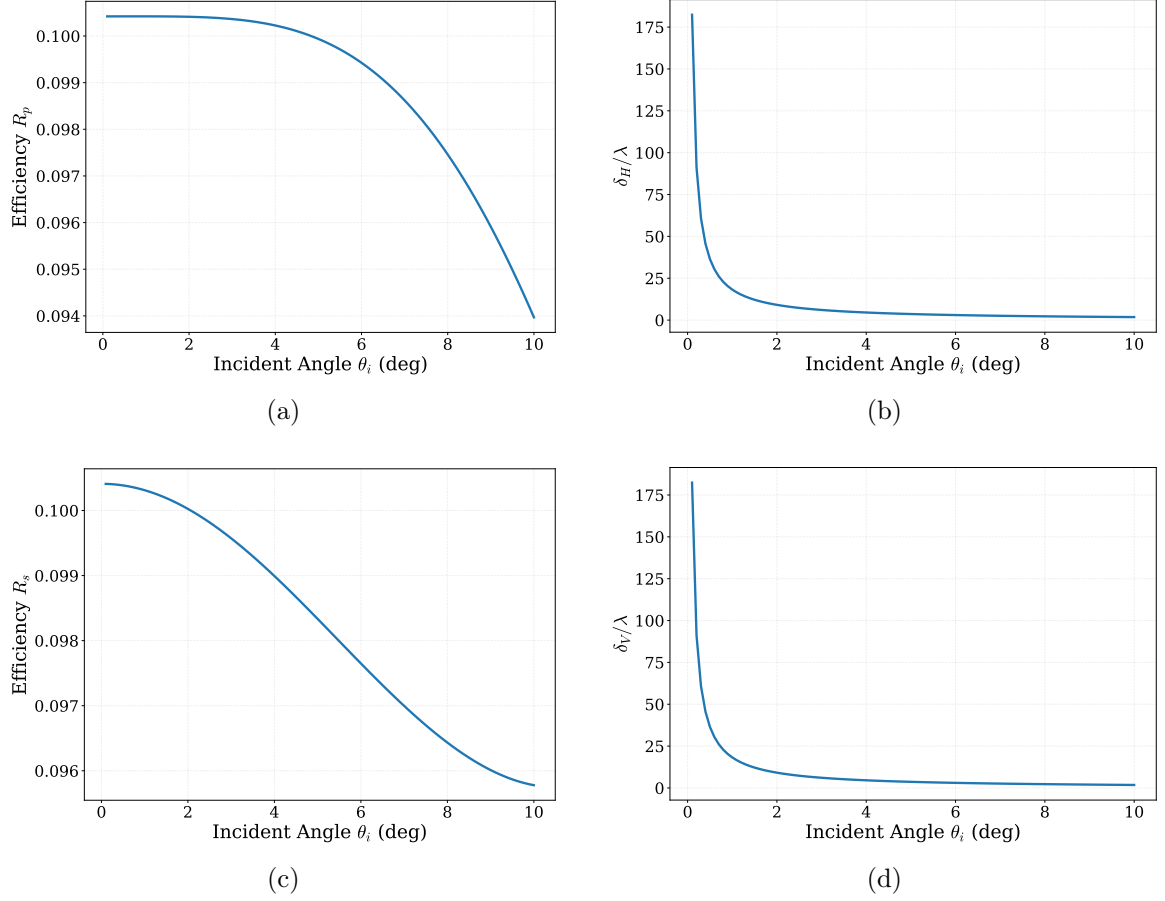


Figure A.5: Additional angular dependence of the normalized SHEL shift and reflectance efficiency for the crystalline Sb_2S_3 metasurface. (a) Reflectance efficiency for H polarization. (b) Normalized SHEL shift for H polarization. (c) Reflectance efficiency for V polarization. (d) Normalized SHEL shift for V polarization.

Bibliography

- [1] Minkyung Kim et al. “Spin Hall Effect of Light: From Fundamentals To Recent Advancements”. In: *Laser & Photonics Reviews* 17.1 (2023). DOI: [10.1002/lpor.202200046](https://doi.org/10.1002/lpor.202200046).
- [2] Jiajun Wang et al. “Shifting Beams at Normal Incidence via Controlling Momentum-Space Geometric Phases”. In: *Nature Communications* 12.1 (2021), p. 6046. DOI: [10.1038/s41467-021-26406-5](https://doi.org/10.1038/s41467-021-26406-5).
- [3] Zhen Hua Ni et al. “Graphene Thickness Determination Using Reflection and Contrast Spectroscopy”. In: *Nano Letters* 7.9 (2007), pp. 2758–2763. DOI: [10.1021/nl071254m](https://doi.org/10.1021/nl071254m).
- [4] Xiang Zhou et al. “Identifying graphene layers via spin Hall effect of light”. In: *Applied Physics Letters* 101.25 (2012), p. 251602. DOI: [10.1063/1.4772947](https://doi.org/10.1063/1.4772947).
- [5] Peng Zhang et al. “High-sensitivity measurement of graphene layers based on the spin Hall effect of light”. In: *Optics Communications* 475 (2020), p. 126175. DOI: [10.1016/j.optcom.2020.126175](https://doi.org/10.1016/j.optcom.2020.126175).
- [6] Yu Zhou et al. “Enhanced Spin Hall Effect of Light at Graphene-Derived Nanostructured Interfaces”. In: *Journal of the Physical Society of Japan* 82.7 (2013), p. 074717. DOI: [10.7566/JPSJ.82.074717](https://doi.org/10.7566/JPSJ.82.074717).
- [7] Andrea Arbabi and Federico Capasso. “Metasurfaces: From fundamentals to applications”. In: *Nature Reviews Materials* 3 (2018), pp. 392–414. DOI: [10.1038/s41578-018-0034-1](https://doi.org/10.1038/s41578-018-0034-1).
- [8] Minkyung Kim et al. “Reaching the highest efficiency of spin Hall effect of light in the near-infrared using all-dielectric metasurfaces”. In: *Nature Communications* 13.2036 (2022). DOI: [10.1038/s41467-022-29771-x](https://doi.org/10.1038/s41467-022-29771-x).
- [9] Kirill Koshelev et al. “Asymmetric metasurfaces with high-Q resonances governed by bound states in the continuum”. In: *Physical Review Letters* 121.19 (2018), p. 193903. DOI: [10.1103/PhysRevLett.121.193903](https://doi.org/10.1103/PhysRevLett.121.193903).
- [10] Jiajun Wang, Lei Shi, and Jian Zi. “Spin Hall Effect of Light via Momentum-Space Topological Vortices around Bound States in the Continuum”. In: *Physical Review Letters* 129.23 (2022), p. 236101. DOI: [10.1103/PhysRevLett.129.236101](https://doi.org/10.1103/PhysRevLett.129.236101).
- [11] Jiajun Wang et al. “Shifting beams at normal incidence via controlling momentum-space geometric phases”. In: *Nature Communications* 12.6046 (2021). DOI: [10.1038/s41467-021-26406-5](https://doi.org/10.1038/s41467-021-26406-5).
- [12] Luis Garza-Soto, Nathan Hagen, and Dorilian Lopez-Mago. “Deciphering Pancharatnam’s discovery of geometric phase: retrospective”. In: *Journal of the Optical Society of America A* 40.5 (2023), pp. 925–931. DOI: [10.1364/JOSAA.485485](https://doi.org/10.1364/JOSAA.485485).
- [13] Eugene Hecht. *Optics*. 5th. Pearson Education, 2016. ISBN: 9780133977226.
- [14] Max Born and Emil Wolf. *Principles of Optics: Electromagnetic Theory of Propagation, Interference and Diffraction of Light*. 7th. Cambridge University Press, 1999. ISBN: 9780521642224.